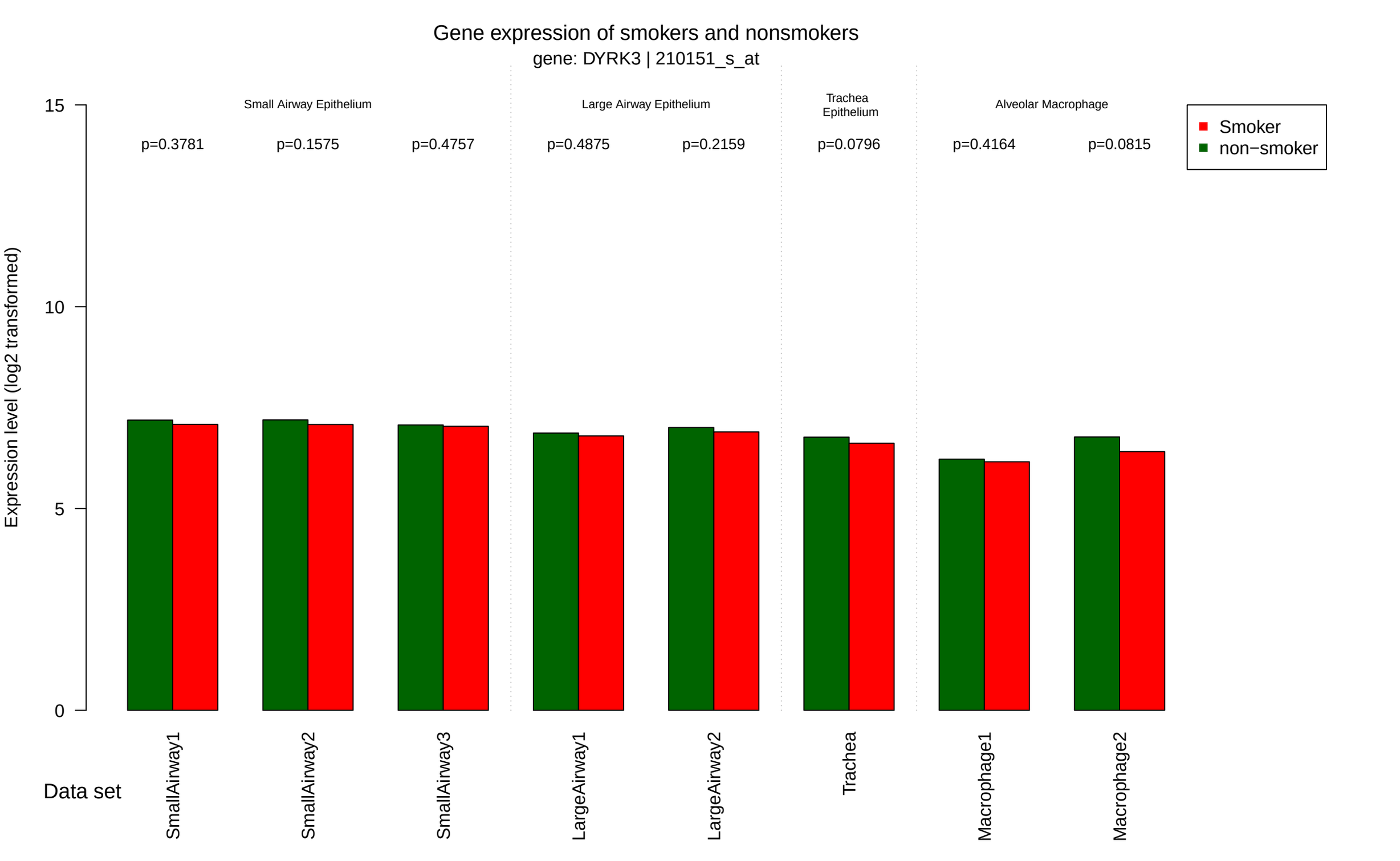
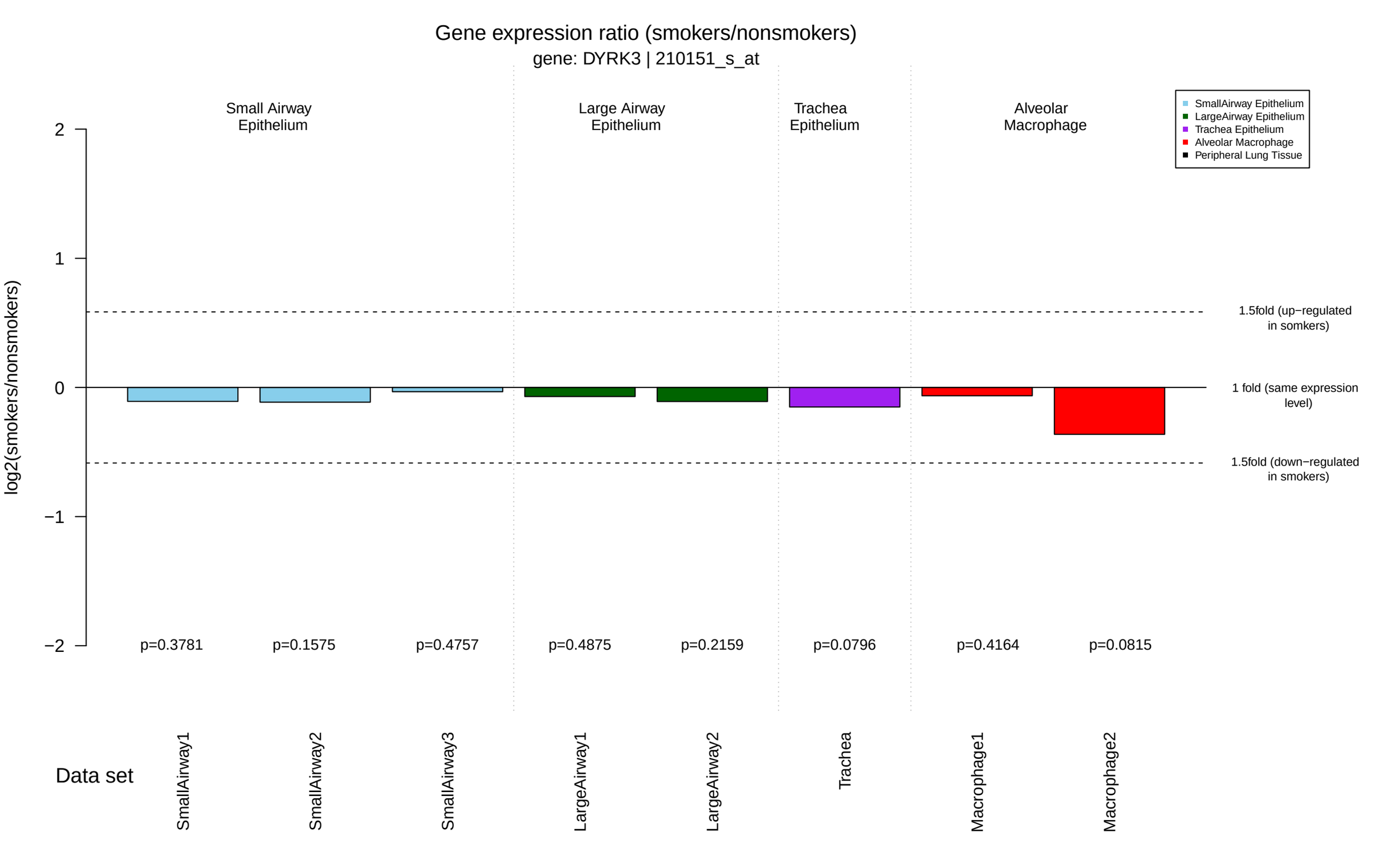
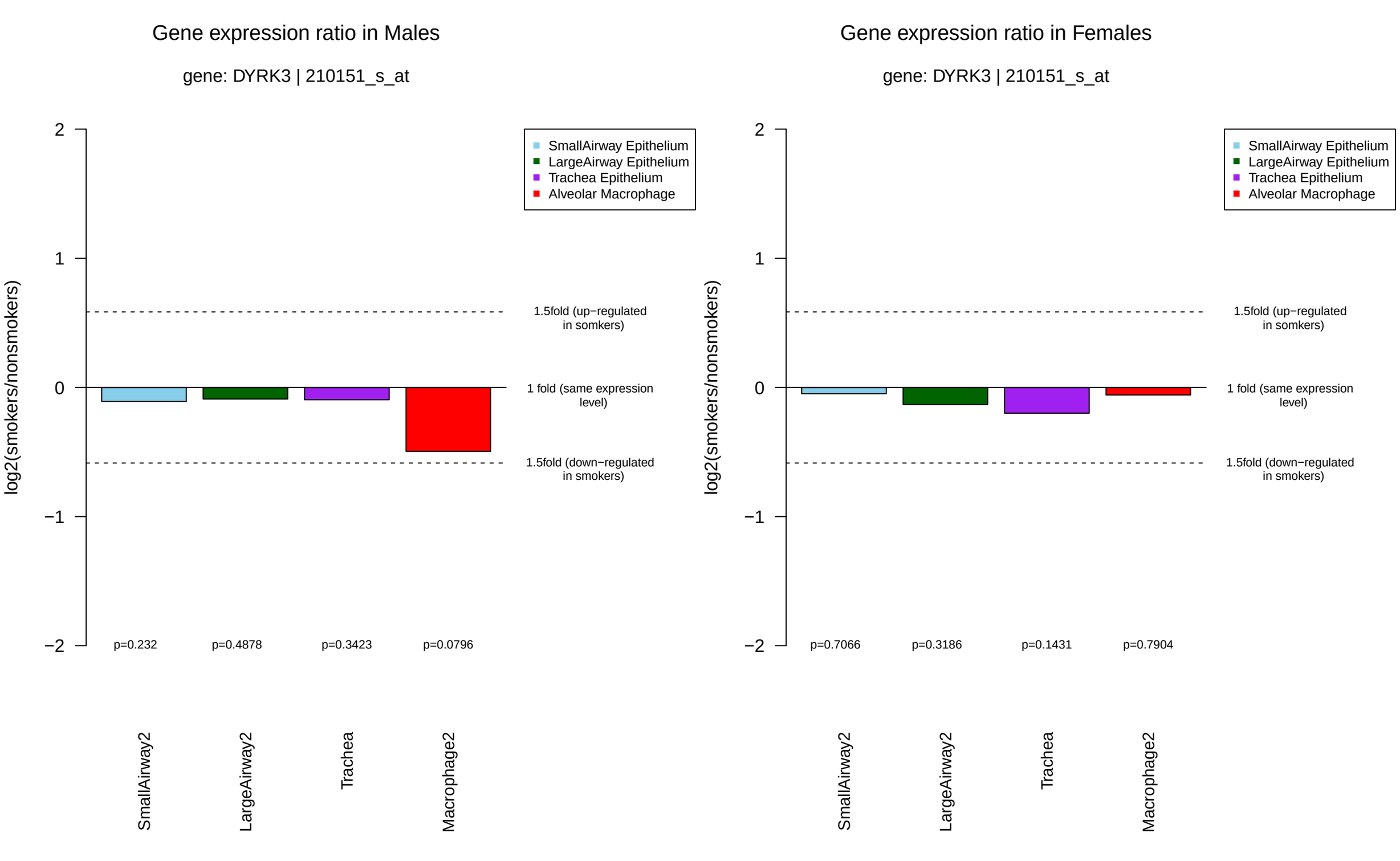
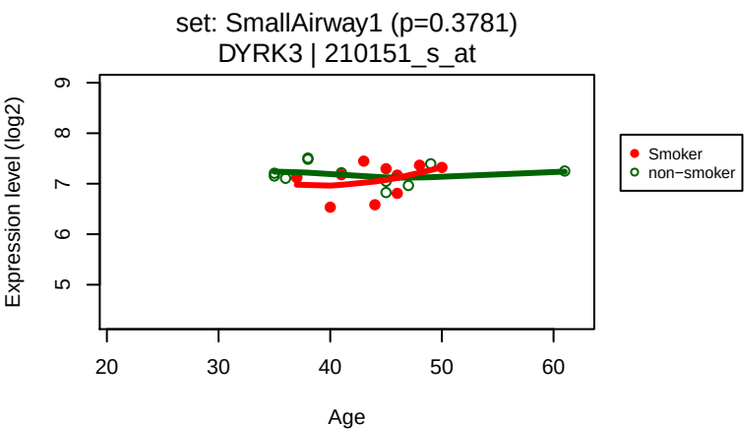
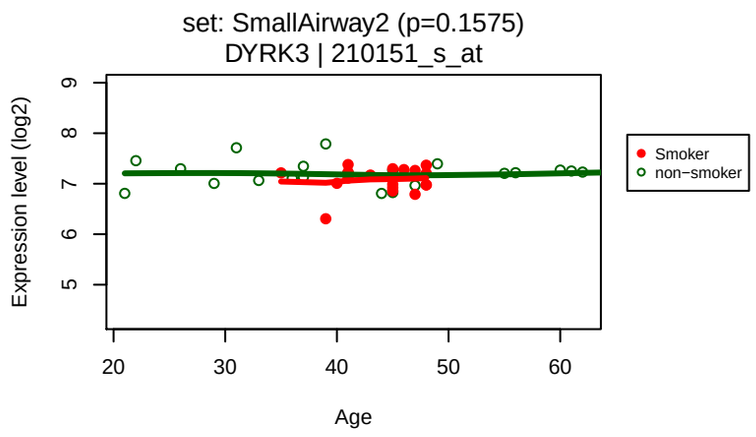
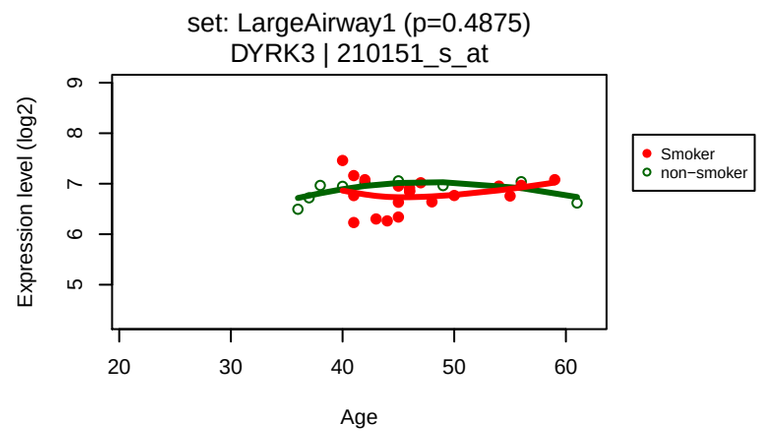

Gene expression of smokers and nonsmokers
gene: DYRK3 | 210151_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3781
p=0.1575
p=0.4757
p=0.4875
p=0.2159
p=0.0796
p=0.4164
p=0.0815
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

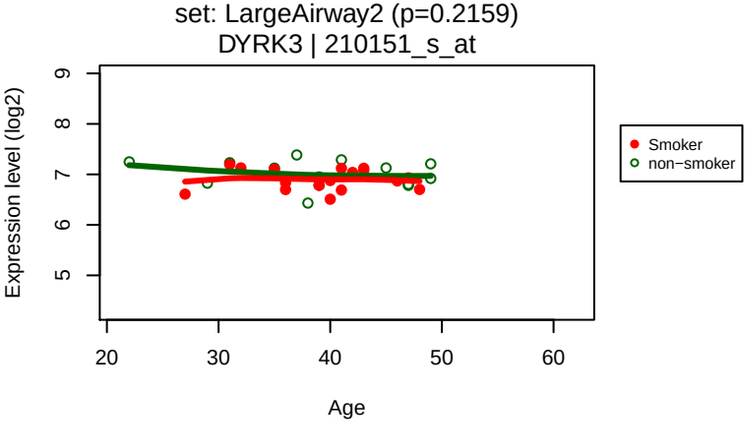

Gene expression ratio (smokers/nonsmokers)
gene: DYRK3 | 210151_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3781
p=0.1575
p=0.4757
p=0.4875
p=0.2159
p=0.0796
p=0.4164
p=0.0815
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

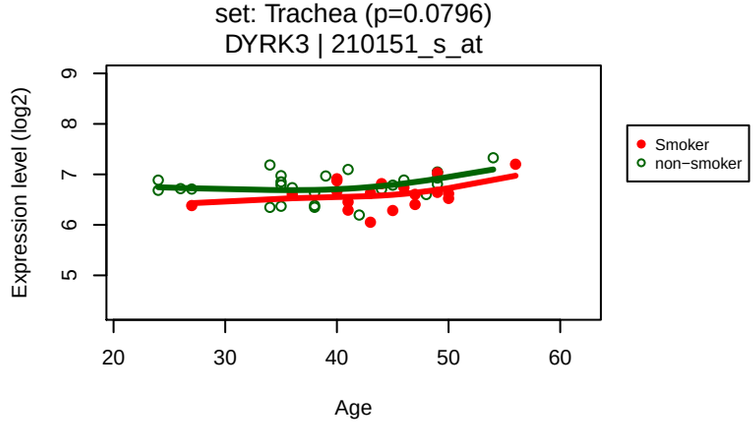

Gene expression ratio in Males
gene: DYRK3 | 210151_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.232
p=0.4878
p=0.3423
p=0.0796
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DYRK3 | 210151_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7066
p=0.3186
p=0.1431
p=0.7904
SmallAirway2
LargeAirway2
Trachea
Macrophage2

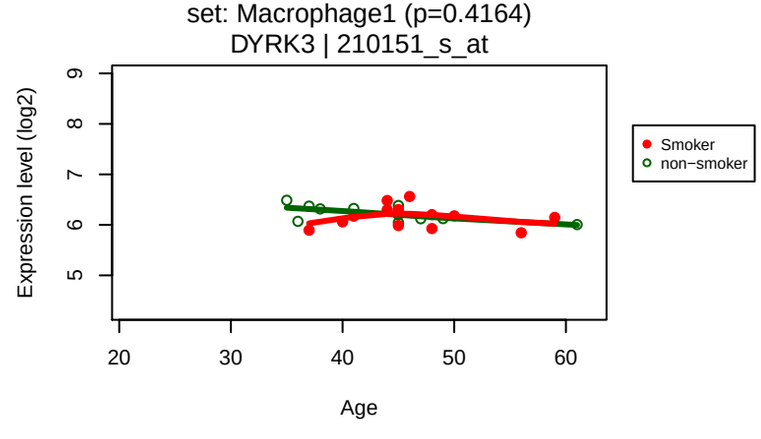

set: SmallAirway1 (p=0.3781)
DYRK3 | 210151_s_at
Expression level (log2)
Age
Smoker
non−smoker

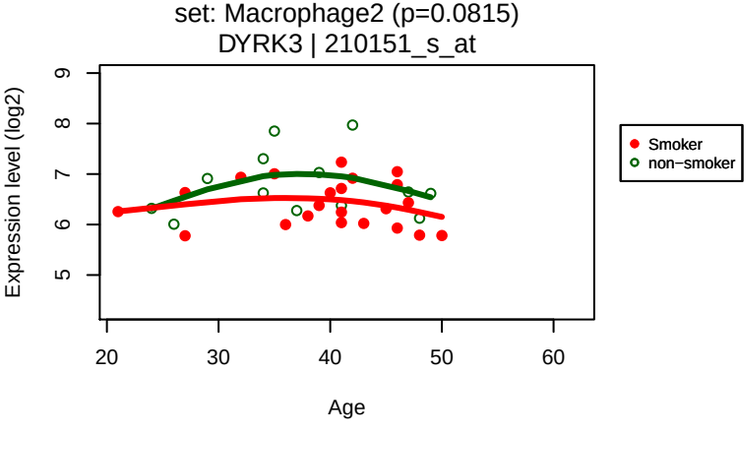

set: SmallAirway2 (p=0.1575)
DYRK3 | 210151_s_at
Expression level (log2)
Age
Smoker
non−smoker

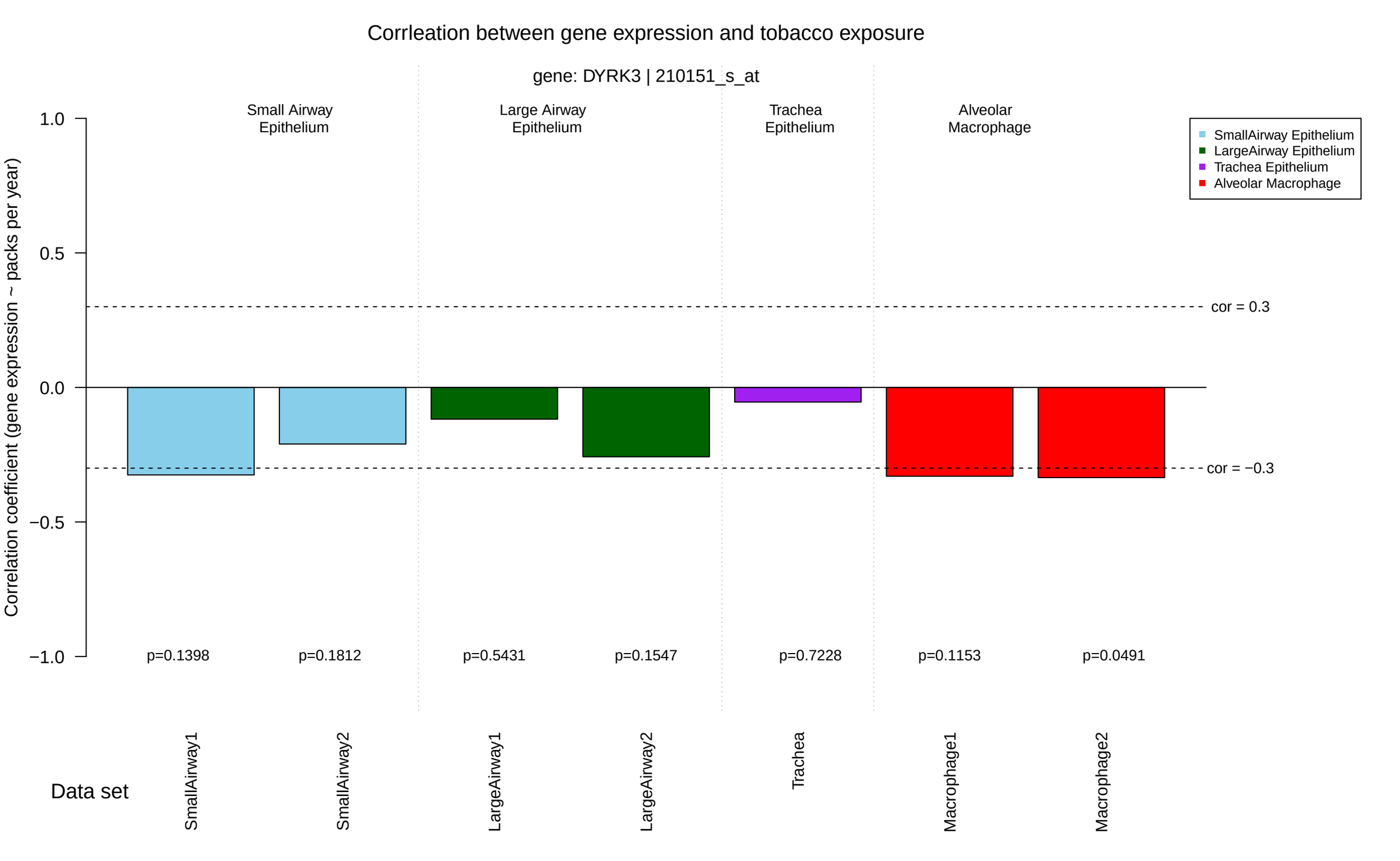

set: LargeAirway1 (p=0.4875)
DYRK3 | 210151_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2159)
DYRK3 | 210151_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0796)
DYRK3 | 210151_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4164)
DYRK3 | 210151_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0815)
DYRK3 | 210151_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DYRK3 | 210151_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1398
p=0.1812
p=0.5431
p=0.1547
p=0.7228
p=0.1153
p=0.0491
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2